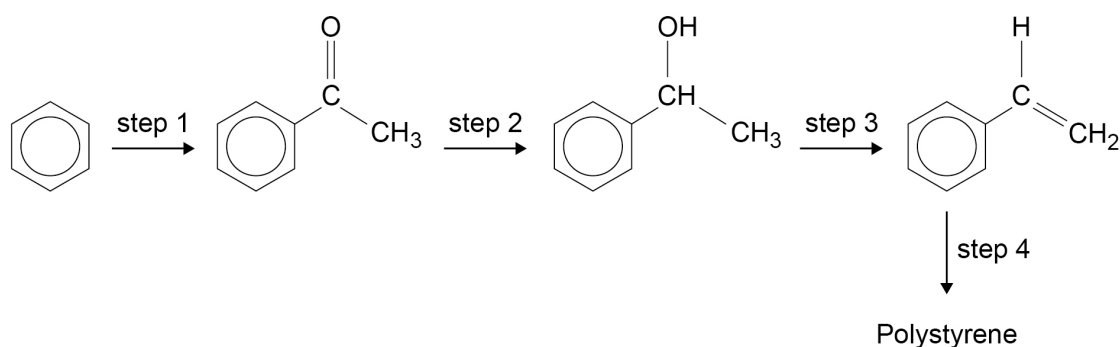


QQ5 Benzene

AQA 2020,2021 OCR 2020

0 6

Polystyrene can be made from benzene in the series of steps shown.

**0 6 . 1**

State the type of reaction in step 1.

Identify the reagent(s) and conditions needed for step 1.

[3 marks]

Type of reaction _____

Reagent(s) _____

Conditions _____

0 6 . 2

State the name of the mechanism for the reaction in step 2.

Identify the inorganic reagent needed for step 2.

Name the organic product of step 2.

[3 marks]

Name of mechanism _____

Inorganic reagent _____

Name of organic product _____



0 6 . 3 The organic product of step **2** is reacted with concentrated sulfuric acid in step **3**.

Outline the mechanism for step **3**.

[3 marks]

0 6 . 4 Draw the repeating unit of polystyrene.

[1 mark]

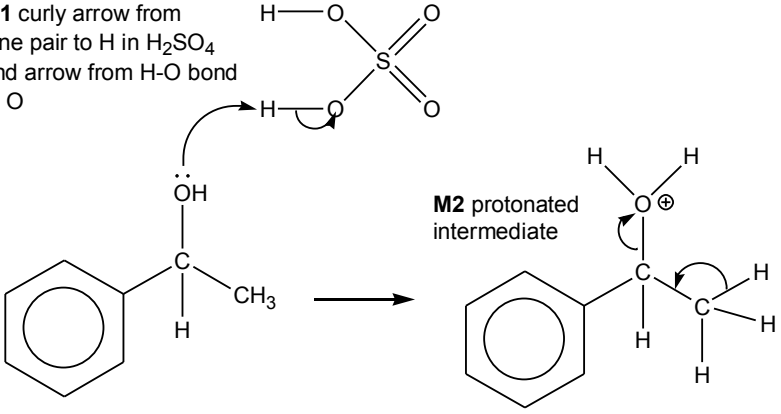
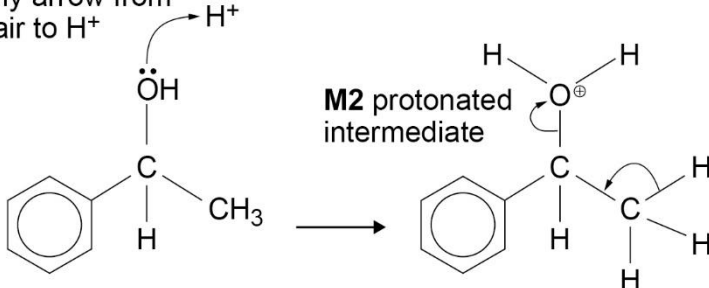
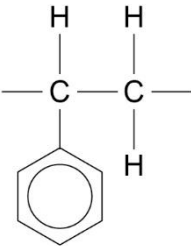
10

Turn over for the next question

Turn over ►



Question	Answers	Additional comments/Guidelines	Mark
06.1	M1 Acylation	Allow electrophilic substitution	1
	M2 CH_3COCl OR Ethanoyl chloride	Allow ethanoic anhydride for M2	1
	M3 AlCl_3 OR Aluminium chloride (mark could be awarded in space for M2)	M3 dependent on M2	1
		Allow Dry/anhydrous for M3 Apply list principle to extra incorrect conditions	
06.2	M1 Nucleophilic addition	Allow LiAlH_4 for M2	1
	M2 NaBH_4	If H_2/Ni stated allow M2 and M3	1
	M3 1-phenyl ethan(-1-)ol	but to score a matching M1 it would have to be Catalytic addition	1

06.3	M1 curly arrow from lone pair to H in H_2SO_4 and arrow from H-O bond to O  M2 protonated intermediate M3 two curly arrows to show loss of water and of H^+ M1 curly arrow from lone pair to H^+  M2 protonated intermediate M3 two curly arrows to show loss of water and of H^+	Penalise M1 for mistakes on structure of H_2SO_4 Allow H^+ attacked in M1 Allow M3 as two steps Allow displayed formulae	3
06.4		Must show trailing bonds Ignore brackets and any use of n Allow C_6H_5 for phenyl group	1

0 4

Kekulé suggested this structure for benzene.



Benzene is now represented by this structure.

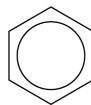


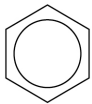
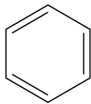
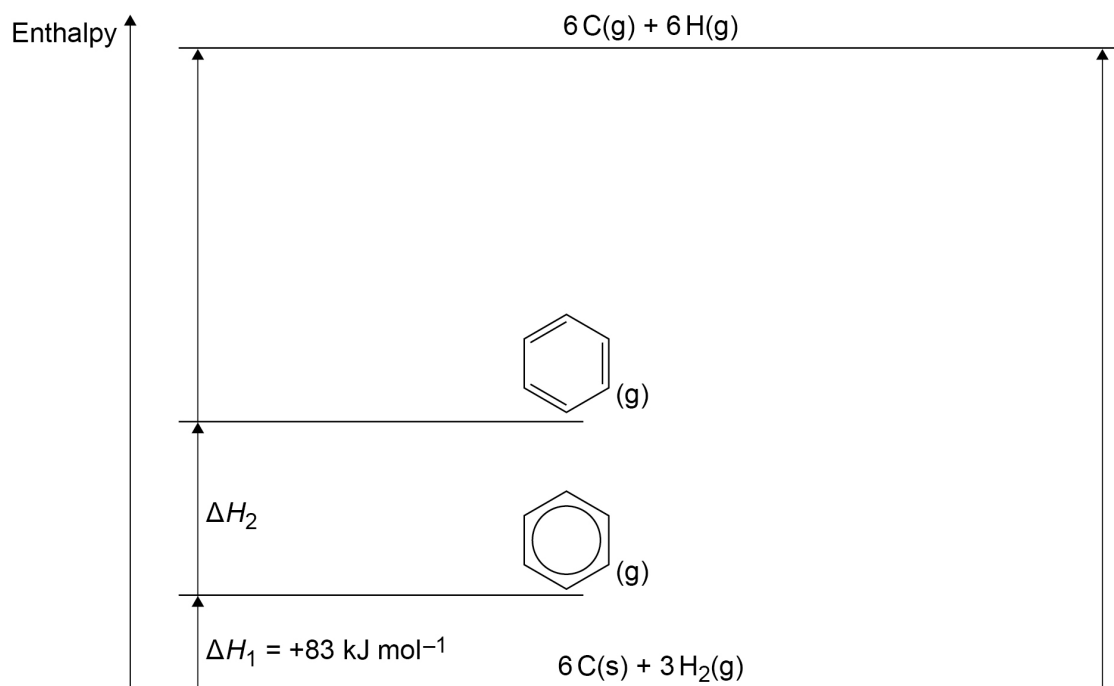
Figure 3 shows the relative stability of  compared to .

Figure 3



0 4 . 1

Use **Figure 3** and the data shown in **Table 1** to calculate ΔH_2 **[3 marks]****Table 1**

	$\Delta H / \text{kJ mol}^{-1}$
Enthalpy of atomisation for carbon	+715
Enthalpy of atomisation for hydrogen	+218
Bond enthalpy (C–C)	+348
Bond enthalpy (C=C)	+612
Bond enthalpy (C–H)	+412

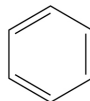
 ΔH_2 _____ kJ mol^{-1}

0 4 . 2

Explain, in terms of structure and bonding, why



is more thermodynamically stable than

**[1 mark]**

Turn over ►

0 4 . 3

A mixture of concentrated nitric acid and concentrated sulfuric acid reacts with benzene.

Figure 4 shows the incomplete mechanism for this reaction.

Name the mechanism.

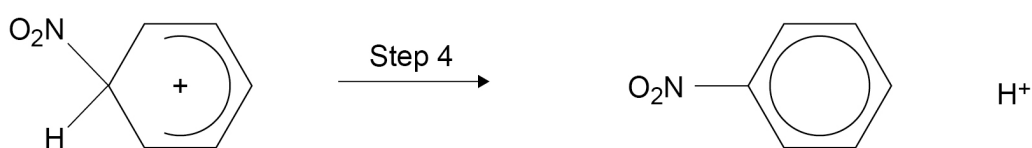
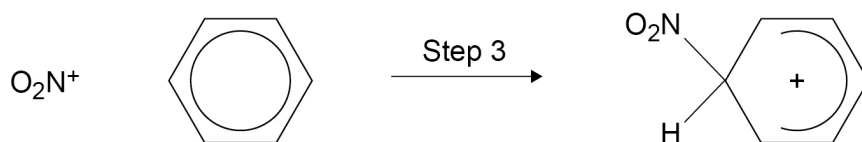
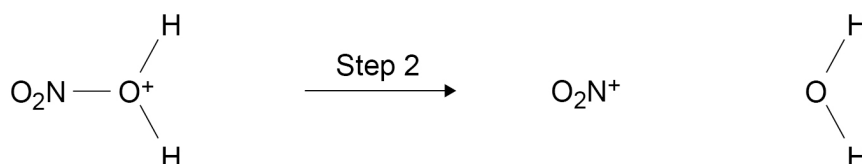
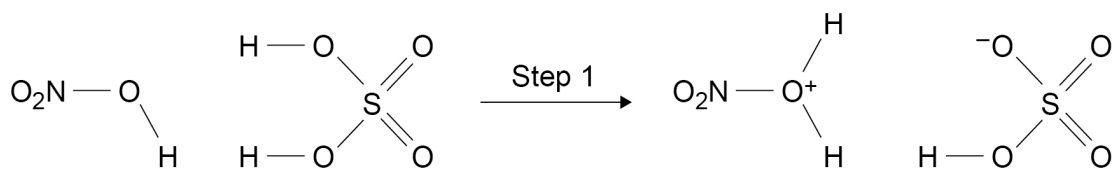
Complete the mechanism in **Figure 4** by adding

- any lone pairs of electrons involved in each step
- **two** curly arrows in step 1
- a curly arrow in step 2
- a curly arrow in step 3
- a curly arrow in step 4.

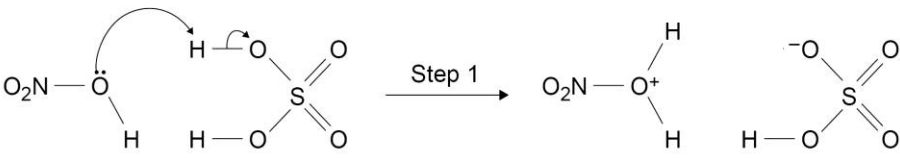
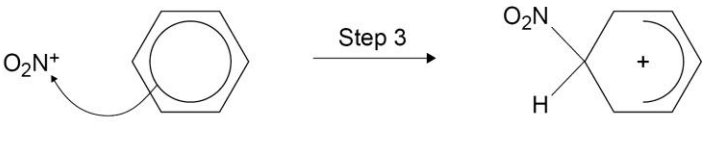
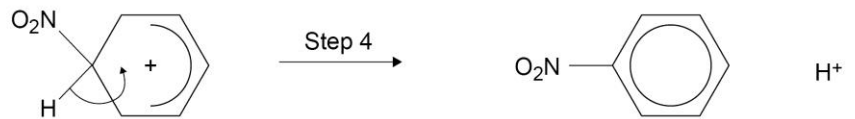
[5 marks]

Name of mechanism _____

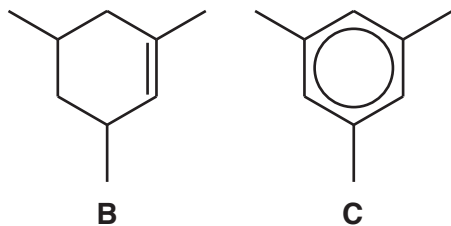
Figure 4



Question	Answers	Additional Comments/Guidelines	Mark
04.1	$(3 \times 612) + (3 \times 348) + (6 \times 412) = 5352$	For LHS	M1
	$(6 \times 715) + (6 \times 218) = 5598$	For RHS	M2
	$\Delta H_2 = M2 - M1 - 83 = +163 \text{ kJ mol}^{-1}$		M3
Question	Answers	Additional Comments/Guidelines	Mark
04.2	(π) electrons delocalised		1

Question	Answers	Additional Comments/Guidelines	Mark
04.3	M1 Electrophilic substitution		1
	M2 for a lone pair and two curly arrows 		1
	M3 for a curly arrow from the bond to the O 		1
	M4 for a curly arrow from inside the hexagon to the N or + on the N 		1
	M5 curly arrow from the bond back into the hexagon 		1

17 Compounds **B** and **C**, shown below, are unsaturated hydrocarbons containing nine carbon atoms.



- (a) Compound **B** reacts with chlorine at room temperature, but compound **C** requires the presence of a halogen carrier.

In both reactions, the organic compound reacts with chlorine in a 1:1 molar ratio.

- (i) Draw the structures of the organic product of each reaction.

Organic product with B	Organic product with C

[2]

- (ii) Explain the relative resistance to chlorination of compound **C** compared with compound **B**.

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[3]

- (iii) Outline the mechanism for the reaction of compound **C** with chlorine.

Show the role of the halogen carrier.

[5]

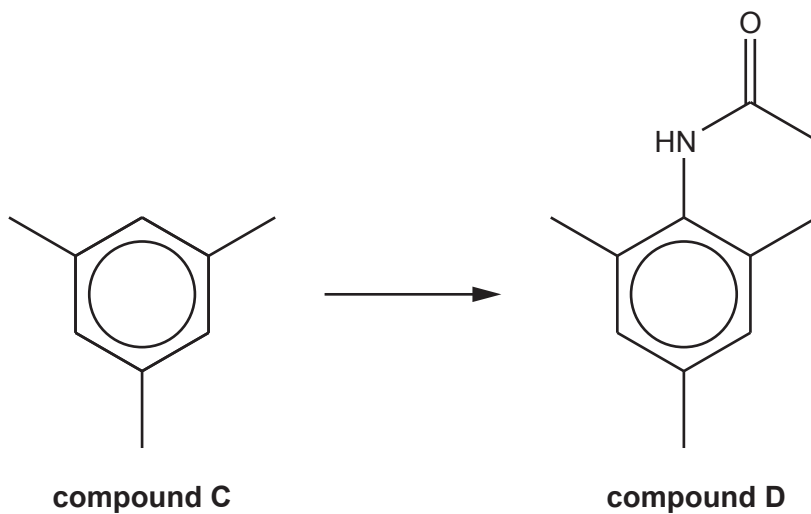
- (b) Compound **C** can be prepared by 'trimerisation' of propanone using concentrated sulfuric acid as a catalyst.

Suggest an equation for this reaction, using **molecular** formulae.

..... [3]

(c) An organic chemist is investigating compound **D** for possible use as a medicine.

The chemist proposes a synthesis of compound **D** from compound **C**.



(i) Predict the number of peaks in the ^{13}C NMR spectra of compounds **C** and **D**.

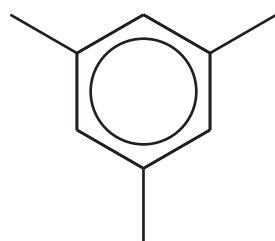
	Compound C	Compound D
Number of peaks		

[2]

(ii) The chemist develops a three-stage synthesis of compound **D** from compound **C**.

Complete the flowchart.

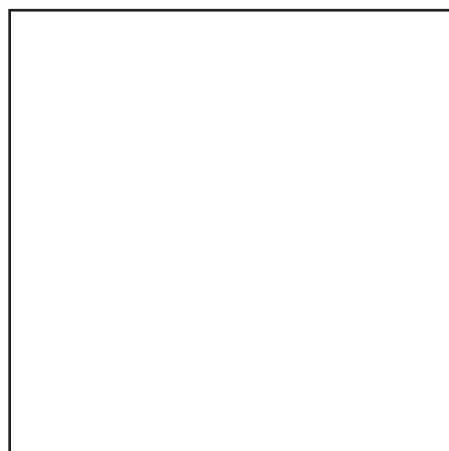
Show structures for organic compounds.



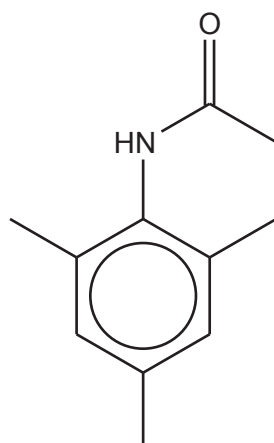
compound C

reagent:

catalyst:

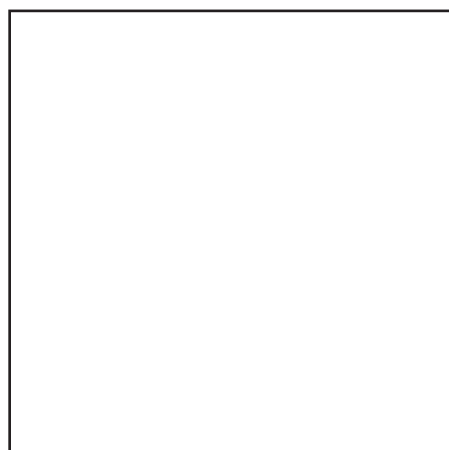


1. Sn + HCl
2. Neutralise

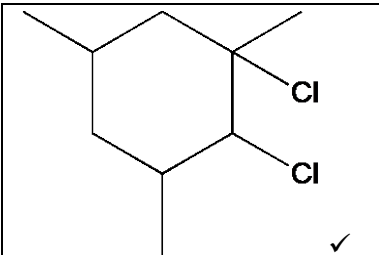
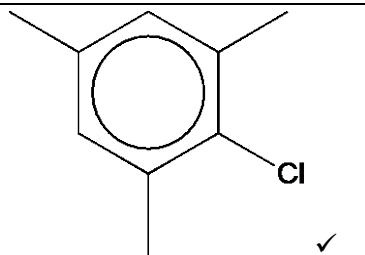


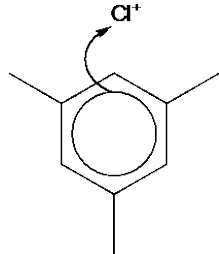
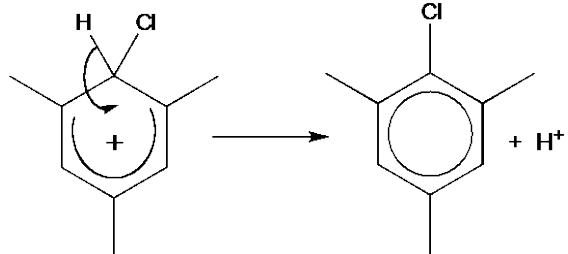
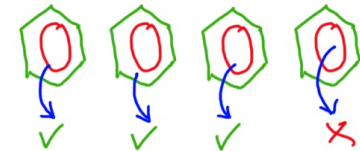
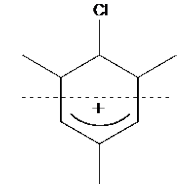
compound D

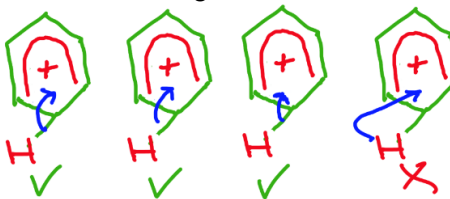
reagent:

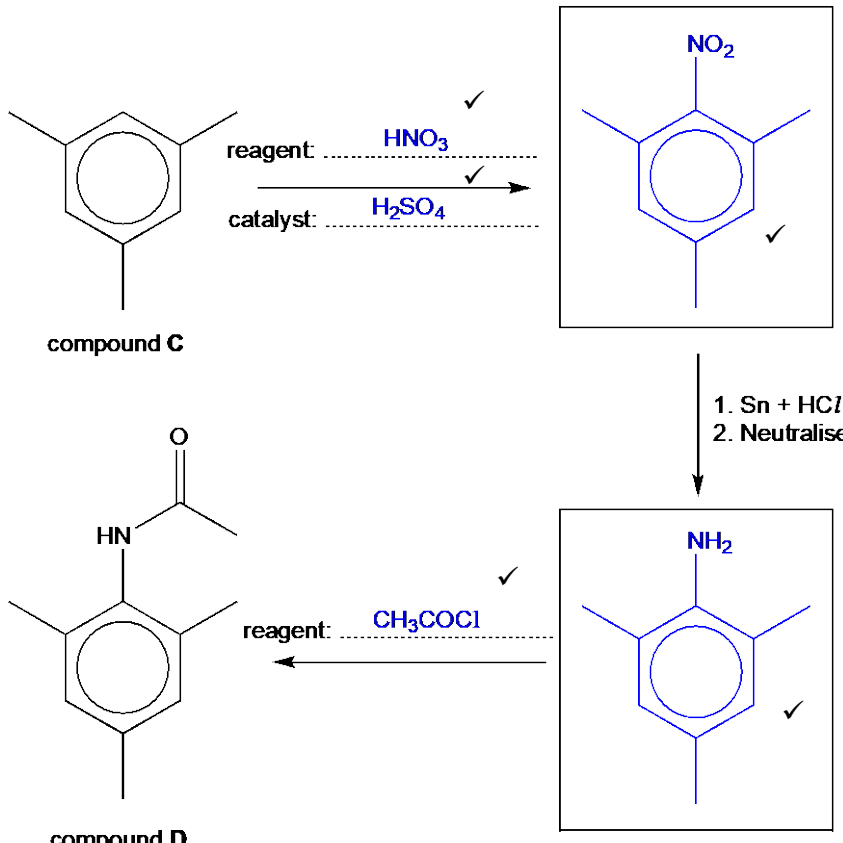


[5]

Question			Answer	Marks	AO element	Guidance
17	(a)	(i)	 Organic product with B  Organic product with C	2	2.5×2	
		(ii)	Reactivity of B in B electrons are localised OR in B π-bond is localised ✓ Reactivity of C in C electrons are delocalised OR In C π-system / ring is delocalised In B, electron density is higher AND B is more susceptible to electrophilic attack OR B attracts/accepts the electrophile/Cl_2 more OR B polarises the electrophile/Cl_2 more ✓ ORA	3	1.1×3	ALLOW labelled diagram to show delocalised system IGNORE charge density IGNORE electronegativity IGNORE B is more reactive/reacts more readily (no reference to electrophile) IGNORE references to electron density spread around the π-ring ALLOW chlorine

Question	Answer	Marks	AO element	Guidance
(iii)	Generation of electrophile $\text{AlCl}_3 + \text{Cl}_2 \rightarrow \text{AlCl}_4^- + \text{Cl}^+ \checkmark$ Attack of Cl^+  Curly arrow from π-bond to Cl^+ $\checkmark$ Intermediate and organic product  Correct intermediate $\checkmark$ Curly arrow from C–H bond to reform π-ring $\checkmark$ Regeneration of catalyst $\text{H}^+ + \text{AlCl}_4^- \rightarrow \text{AlCl}_3 + \text{HCl} \checkmark$	5	1.2 1.2 2.5 1.2 1.2	ANNOTATE ANSWER WITH TICKS AND CROSSES ALLOW $\text{FeCl}_3 + \text{Cl}_2 \rightarrow \text{FeCl}_4^- + \text{Cl}^+$ ALLOW use of Fe NOTE: curly arrows can be straight, snake-like, etc. but NOT double-headed or half-headed arrows 1st curly arrow must start from, OR close to, circle of benzene ring  DO NOT ALLOW following intermediate:  π-ring must cover 4 of the 6 sides of the benzene ring AND correct orientation, <i>i.e.</i> gap towards C–Cl ALLOW + sign anywhere inside the 'hexagon' of the intermediate.

Question			Answer	Marks	AO element	Guidance						
						IGNORE partial charges on the chlorine in the intermediate DO NOT ALLOW mark for intermediate if any CH₃ is missing Curly arrow must start from, OR be traced back to, any part of C-H bond and go inside the 'hexagon'  ALLOW use of AlCl₄⁻ in the mechanism ALLOW ECF for regeneration of an incorrect metal chloride catalyst e.g. AgCl₃						
	(b)		3C₃H₆O → C₉H₁₂ + 3H₂O molecular formulae of C₃H₆O AND C₉H₁₂ ✓ H₂O as by-product ✓ correct balanced equation ✓	3		2.6 2.5 2.6						
	(c)	(i)	<table><tr><td></td><td>Compound C</td><td>Compound D</td></tr><tr><td>Number of peaks</td><td>3 ✓</td><td>8 ✓</td></tr></table>		Compound C	Compound D	Number of peaks	3 ✓	8 ✓	2	3.2	
	Compound C	Compound D										
Number of peaks	3 ✓	8 ✓										

Question	Answer	Marks	AO element	Guidance
(ii)	 compound C reagent: HNO_3 catalyst: H_2SO_4 1. Sn + HCl 2. Neutralise reagent: CH_3COCl compound D	5	3.2×5	ALLOW any combination of skeletal OR structural OR displayed formula as long as unambiguous IGNORE names for organic intermediates (question asks for structures) ALLOW names of reagents and catalyst Around top arrow, ALLOW 1 of 2 marks if HNO_3 and H_2SO_4 swapped. i.e. reagent: H_2SO_4 catalyst: HNO_3 IGNORE references to concentration ALLOW $(\text{CH}_3\text{CO})_2\text{O}$ for left arrow IGNORE CH_3COOH IGNORE acyl chloride DO NOT ALLOW $\text{AlCl}_3/\text{FeCl}_3/\text{Fe4}$